

Machine Learning Engineer

Freight Rate Prediction Assessment Report

Candidate: Rayen Oueslati · Target: predict **posted_rate** for truck loads

1. Objective & Framing

This report documents the approach used for Spotter’s freight-rate assessment. Labeled development data covers January–October 2025 (48,000 loads). Unlabeled validation covers November–December 2025 (12,000 loads). Because freight markets are non-stationary, the problem is treated as a **chronological forecasting task** rather than a shuffled regression.

2. Train / Validation Split

Primary holdout protocol used for model selection and reporting:

- **Train:** 2025-01-01 → 2025-09-30 (extreme rates winsorized at the 99.7th percentile while fitting).
- **Holdout:** 2025-10-01 → 2025-10-31 (evaluation remains unclipped).
- **Early stopping:** last ~10% of the training window by date.
- **Final model:** refit on all labeled months (Jan–Oct), then score validation and the December chart inputs.

This mirrors deployment: every validation load is strictly later than every labeled observation. Random k-fold CV would leak future market conditions into earlier predictions.

Metric	Value
Holdout period	October 2025
Backend	xgboost_cuda
MAE	USD 108.79
RMSE	USD 649.49
MAPE	5.15%
Train / test rows	43,023 / 4,853
Blend	85% XGBoost + 15% Huber

October holdout summary: MAE = **USD 108.79**, RMSE = **USD 649.49**, MAPE = **5.15%**.

3. Exploratory Findings

- **Distance dominates price** — correlation with posted_rate ≈ 0.91 ; median $\approx$ USD 2.15 / mile.
- **Equipment premium:** Reefer > Flatbed > Dry Van on a per-mile basis.
- **Market index** is nearly constant within a day and strongly seasonal across the year (daily market $\leftrightarrow$ RPM ≈ 0.64).
- **Quote signal** varies load-to-load; distance $\times$ quote is highly informative.
- 64 cities with stable coordinates (one lat/lon pair per city).

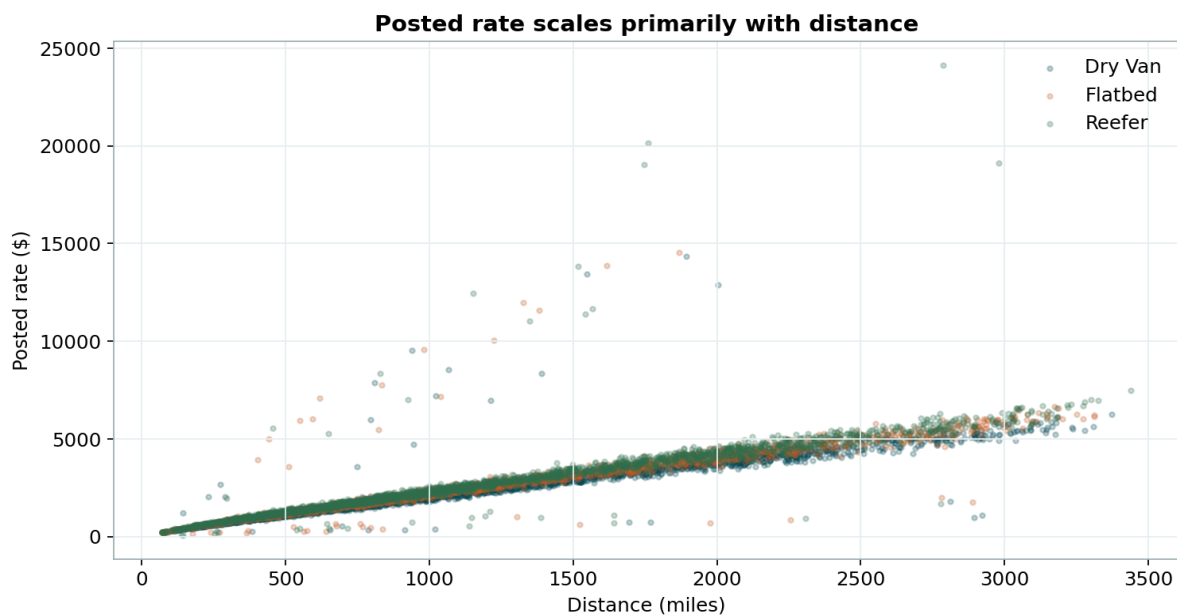


Figure 1. Posted rate vs distance by equipment.

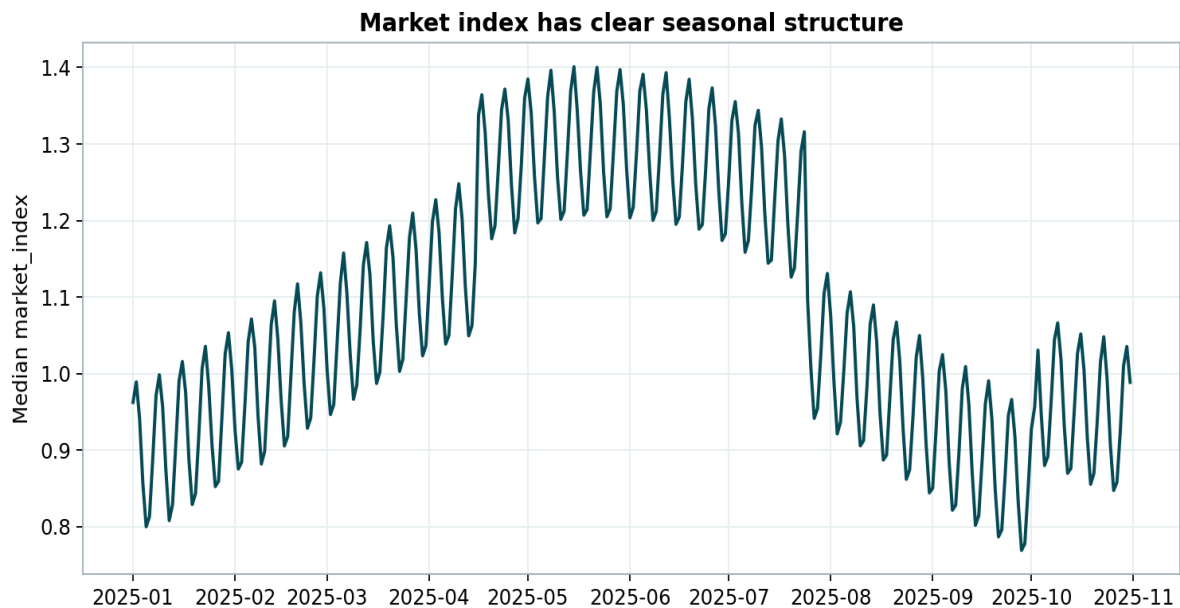


Figure 2. Seasonal structure in market_index.

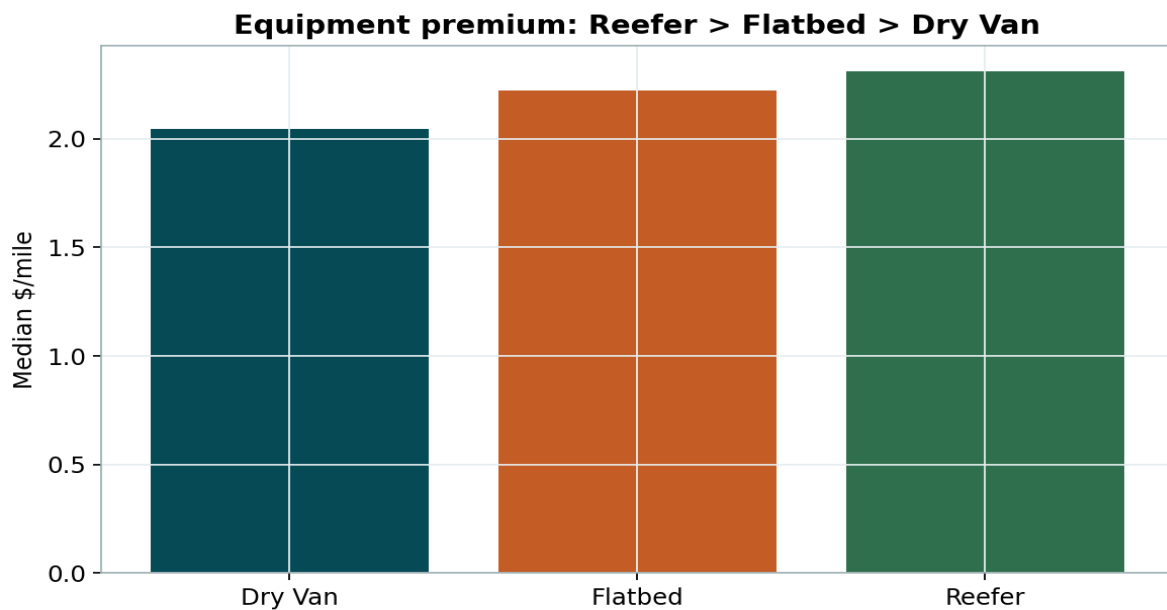


Figure 3. Median USD/mile by equipment.

4. Data-Quality Issues & Remediation

- **Negative weights (~292 rows):** treated as sign errors → absolute value.
- **Missing weight (~300) and market_index (~374):** median imputation fit on the training fold only.
- **Extreme posted rates:** winsorized at the 99.7th percentile during fitting; holdout evaluation unclipped.

- **December template gap:** no lat/lon, market_index, or quote_signal. Coordinates looked up from training cities; market/quote forecasted from historical daily patterns.

5. Modeling Approach

Selected model: **GPU-accelerated XGBoost** (hist + CUDA) trained on **log1p(posted_rate)** with an MAE objective, blended **85% / 15%** with a robust **Huber** linear model on economically motivated interactions (distance × market, distance × quote, equipment slope terms).

Why this family

- Tabular mix of continuous and low-cardinality categorical features — gradient boosting is a strong default.
- Log target stabilizes heteroscedasticity (absolute error grows with distance).
- MAE objective matches preference for typical absolute dollar error.
- CUDA hist keeps iteration cost low; CPU HistGradientBoosting remains as fallback.

6. December Fixed-Lane Forecast

The scorer holds pickup, delivery, distance, equipment, and weight fixed (Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb). Only date changes. Market and quote series are forecasted from seasonality and recent residuals so the chart reflects temporal dynamics rather than a flat constant mile-rate.

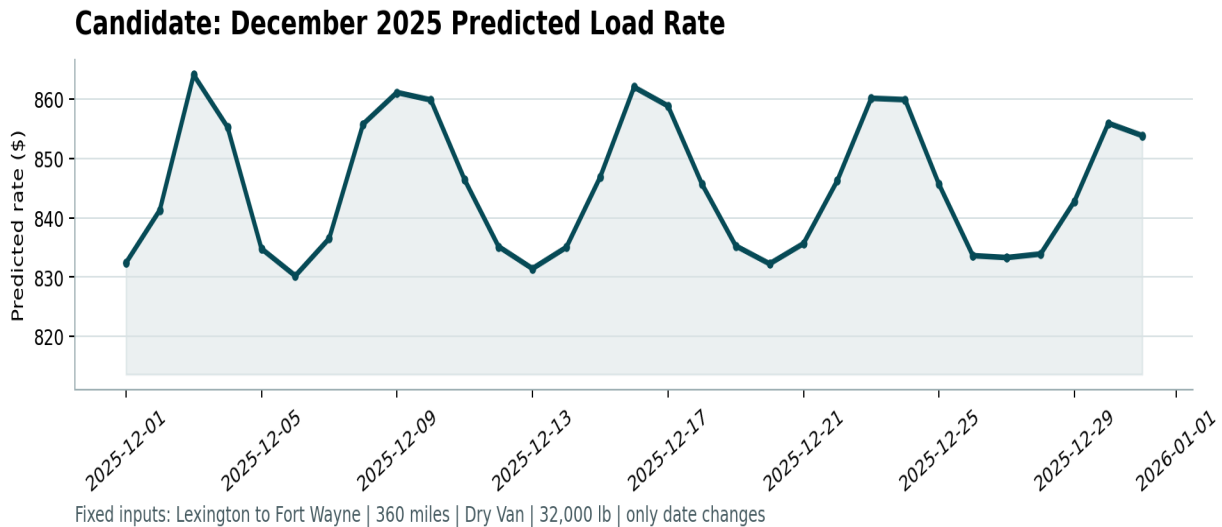


Figure 4. Fixed December prediction chart produced by the provided score.py.

7. How to Reproduce

```
python -m pip install -r requirements.txt
python scripts/eda.py
python train_predict.py
python score.py --predictions validation_predictions.csv --december-predictions
data/december_chart_inputs.csv
```

8. Deliverables

- GitHub repository with solution code, dependencies, and run instructions.
- validation_predictions.csv — exactly load_id, predicted_rate for 12,000 loads.
- This PDF report — chronological split rationale + official December chart.